Project Cover Sheet

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Title:

One Core to Many: Parallel Data-Computing Solutions to the 2023 NYC Yellow-Taxi Trips

Highlights:

- Of 38,310,226 yellow-taxi rows in 2023, 34,893,980 remain after seven rules (8.92% removed). 89.8% of dated pickups start in Manhattan. The busiest cell is Thursday at 18:00 (405,828 trips).
- Silhouette peaks at k = 2 (0.550), which separates JFK from the city. The reported grouping is k = 4 (silhouette 0.312): weekday morning, weekday afternoon, weekend, and a JFK group.
- Zone 132, JFK Airport, is 98.3% of that group (10,903 of 11,092 rows). Its mean trip is 16.1 miles, 42.0 minutes, and $63.69. LaGuardia is not in the group.
- On this 4-core host the process-pool ETL falls from 52.135 s to 13.658 s at four workers (speedup 3.82) and rises to 14.266 s at eight. Four threads take 27.423 s. One chunk of all twelve files takes 49.668 s.
- The k-means kernel falls from 6.428 s to 1.611 s (speedup 3.99). FP-Growth does not speed up (S(4) = 0.89) because the prefix tree is built before the pool starts.

List of Deliverables

File Name	Description
Project_cover_sheet.pdf	Cover sheet for this project
Project_report.pdf	Main project report
Nyc.pptx	Presentation Slides